

Sajeevan Veeriah

ROBOTICS & AUTONOMOUS SYSTEMS ENGINEER

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PROFESSIONAL PROFILE

Robotics and autonomous-systems engineer with the breadth of a complete multidisciplinary engineer — fluent across mechanical design and CAD, electrical and electromechanical systems, electronics and embedded firmware, industrial controls and automation, robotics and autonomy, and software, data and applied AI/ML. On the autonomy side I build and validate the full stack: perception, SLAM and state estimation through motion planning, control and embedded firmware to integration, field testing and documented handover. Recent delivery spans an end-to-end ROS 2 autonomous rover, vehicle software and ADAS validation with CAN tooling, GMP smart-factory automation, and end-to-end IoT with ML-driven monitoring and anomaly detection. Comfortable identifying which engineering layer a problem lives in and resolving it across whatever disciplines that takes.

CORE TECHNICAL SKILLS — FULL ENGINEERING STACK

Robotics & Autonomy	ROS 2 (Humble), Nav2, MoveIt 2, Gazebo, RViz, SLAM, A* path planning, motion control, PID tuning, autonomous navigation.
State Estimation & Fusion	Kalman / EKF state estimation, odometry, sensor fusion, LiDAR, IMU, depth cameras, localisation and mapping.
Perception & Computer Vision	OpenCV, image processing, feature and object detection, visual odometry, camera/LiDAR data pipelines.
AI / ML & Data	Applied machine learning, ML fundamentals, anomaly detection, data analysis & visualisation, MATLAB / Simulink, sensor-data pipelines, InfluxDB, Grafana.
Embedded & Real-time	Embedded C/C++, ESP32 / ESP32-S3, STM32, FreeRTOS, UART, I2C, SPI, CAN, ADC, PWM, signal conditioning, board bring-up.
Software & Tooling	Python, C, C++, JavaScript, TypeScript, Node.js, REST APIs, MQTT, Linux, Git, JIRA, test automation, CI workflows.
Automotive & Validation	Vector CANoe, CANalyzer, CAN bus, CAN FD, ADAS validation, OTA regression testing, vehicle instrumentation, fault isolation.
Hardware, CAD & Electromechanical	PCB design (Altium, KiCad), SolidWorks, Fusion 360, motors & drives, VFDs, actuators, sensors, instrumentation, 3D printing.
Controls, Automation & QA	PLC (IEC 61131-3), Siemens TIA Portal, WinCC, PCS 7, iFIX, PVI+, HMI/SCADA, MES, Modbus, Profinet; GMP/GAMP 5, ISO 13849, IEC 62443, FMEA, FAT/SAT.

SELECTED ROBOTICS & ENGINEERING PROJECTS

- Autonomous Navigation Rover (ROS 2).** Built a complete autonomous navigation stack on ROS 2 Humble with Gazebo/RViz simulation, LiDAR-based SLAM, A* path planning, Kalman/EKF state estimation and odometry/IMU sensor fusion for robust localisation and obstacle-aware navigation.
- ESP32 Clinical Ataxia Assessment Device.** Designed embedded hardware and firmware on ESP32 with IMU, ToF, Hall-effect and magnetometer sensing, real-time signal acquisition, MATLAB data logging and measurement validation against clinical references.
- IoT GPS & Environmental Monitoring Platform.** Delivered ESP32 GPS and environmental sensing over LoRaWAN with MQTT, ChirpStack, InfluxDB and Grafana — firmware, device provisioning, telemetry dashboards and field deployment.

PROFESSIONAL EXPERIENCE

- Automation & Controls Engineer · JAG Process Solutions Pty Ltd** Brunswick, VIC · Jan 2026 – Jun 2026
- Delivered control, integration and smart-factory engineering for pharmaceutical, biotech and food clients under GMP — control logic, HMI/process visualisation and batch execution across plants, skids and packaged units.
 - Executed an iFIX-to-PVI+ SCADA migration, converting application content and verifying functional behaviour against the validated system; produced FAT, SAT, commissioning, qualification and handover documentation.
 - Integrated field devices, sensors, drives and process equipment with control logic and production data flows, with attention to diagnostics, safety, usability and data integrity.
- Product Development Test Engineer (Contract) · Ford Motor Company via Invenio** Little River, VIC · Oct 2025 – Jan 2026
- Validated vehicle software integration and ADAS features across the T6 Ranger and Everest programmes — feature-vehicle, breadboard and regression testing for readiness milestones, OTA updates and sign-off evidence.
 - Instrumented test vehicles and ran structured drives, capturing and analysing CAN-bus data with Vector CANoe and CANalyzer for fault isolation and evidence-based defect reporting.

Technical Assistant · ABMARC

South Geelong, VIC · Jul 2024 – Aug 2025

- Conducted vehicle emissions and compliance testing against ADR and EURO standards, operating test equipment, instrumentation and data-acquisition systems for repeatable, auditable and technically defensible results.
- Prepared compliance, QA and regulatory documentation for certification and audits; analysed test data to identify deviations, trends and evidence gaps.

Consultant Engineer, IoT & Projects Administrator · DuxTel Pty Ltd

Greater Geelong, VIC · Feb 2024 – Aug 2024

- Designed and deployed end-to-end IoT systems linking sensors, embedded devices, gateways, cloud platforms and dashboards, with real-time ingestion, monitoring and visualisation using InfluxDB and Grafana.
- Commissioned IoT devices onto ChirpStack-based LoRaWAN networks and built an ESP32 GPS and environmental-sensing workflow — firmware configuration, device provisioning, testing and deployment.

Earlier Manufacturing & Engineering Experience · IDL, Carbon Revolution & Thornton Engineering Geelong / Corio, VIC · 2018 – 2024

- Manufacturing operations, equipment operation, QA, engineering-drawing review, ITP/MDR documentation, material traceability, safety and first-level troubleshooting across food/beverage, carbon-fibre and structural-steel manufacturing.

EDUCATION

Bachelor of Mechatronics Engineering (Honours)

Completed Jul 2025

- Deakin University — Geelong, VIC, Australia
- Graduated with Distinction (Honours)
- Capstone: ROS 2 autonomous navigation rover (SLAM, EKF, A* path planning)

Higher National Diploma — Mechatronics, Robotics & Automation Engineering

Completed Sep 2016

- Cardiff Metropolitan University — United Kingdom
- Awarded with Distinction

MEMBERSHIPS, CERTIFICATIONS & LANGUAGES

Membership: Member, Engineers Australia

Licence: Current Victorian Driver Licence

Certifications: Lean Six Sigma Foundation; JIRA & Agile; KAIZEN; Industrial Automation & IoT Foundations

Languages: English (Native), Tamil (Native), Sinhala (Native)

REFERENCES

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